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Diagnosis of Alzheimer's Disease Using Boosting Classification Algorithms

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***Abstract*— Alzheimer's Disease (AD) is a progressive degenerative disorder of the brain that impacts memory, cognition, and, ultimately, the ability to carry out daily activities. There is presently no cure for Alzheimer's Disease. However, there are available treatments to manage symptoms and slow their advancement. This research conducted a comprehensive study to diagnose AD using four different categorization methods. These methods included XGBoost, GradientBoost, AdaBoost, and voting classification algorithms. To carry out the examination, a high-quality dataset was obtained from the collection of machine learning data of the prestigious University of California. This dataset was carefully selected to ensure accurate and reliable results. The analysis of the collected data revealed some interesting findings. XGBoost exhibited an accuracy rate of 85% in diagnosing Alzheimer's Disease. ADABoost also performed, achieving an accuracy rate of 75%. GradientBoost, similarly, obtained an accuracy rate of 85%.**

Additionally, the voting classification algorithms showed promise, attaining an accuracy rate of 80%. All these accuracy rates were obtained by implementing a 5-fold cross-validation methodology, which ensured robust and unbiased results. This research contributes to the field of AD diagnosis by providing insights into the effectiveness of different categorization methods.

Keywords — Alzheimer's Disease, XGBoost, ADABoost, GradientBoost.

I. INTRODUCTION

Alzheimer's Disease (AD) is a degenerative brain condition that causes memory loss, cognitive decline, and changes in behavior. It is the predominant type of dementia, responsible for 60-80% of cases, and primarily affects older individuals. As time passes, AD progressively deteriorates, resulting in diminished cognitive skills and difficulty carrying out everyday activities. Detecting AD at an early stage is crucial because it enables the implementation of timely interventions and treatment strategies [1].

AD is a highly challenging disease to treat. It significantly disrupts the lives of elderly individuals and their families, causing considerable hardship. The study of AD diagnosis has experienced significant growth in recent years. A Convolution Neural Network (CNN) architecture is proposed in [2] to diagnose AD using dynamic medical images and structural reparameterization. The 2D reparameterization CNN architecture is designed for classifying Alzheimer's disease. In [3], researchers utilized different machine learning algorithms and strategies to distinguish individuals as either patients or in a healthy state.

In addition, they employed principal component analysis and feature selection approaches, making use of data gathered from neuropsychological assessments, cognitive exercises, or a combination of both. Various machine learning algorithms were employed in [4] to diagnose Alzheimer's patients relative to individuals with healthy brains. Among the classifiers mentioned, the Bagged Ensemble algorithm stood out with exceptional performance, achieving a 90.3% accuracy rate. A unique study in [5] focuses on diagnosing patients using magnetic resonance imaging scans. This research utilizes deep learning techniques to achieve accurate diagnoses.

Additionally, a separate framework is implemented to monitor patients' everyday activities through body-worn inertial sensors actively. In [6], an AI model is suggested, which is a fusion of transfer learning and machine learning algorithms that are permutation-based. It utilizes SVM, Naïve base, XGBoost, and a voting classifier. The model is designed in two primary phases and is applied to a larger dataset. The detection of AD can be improved by using various machine learning techniques on a combination of different biomarkers, as proposed in [7]. The results from the Decision Trees (DT) algorithm showed an accuracy of only 79%.

On the other hand, the XG Boost algorithm, SVM, and Rainforest classifier achieved accuracies of 80%, 81%, and 81% respectively. To further enhance accuracy, the data set was subjected to complex and soft voting classifiers, which yielded 84% and 86% accuracy, respectively. The proposed method [8] can be applied without the need for nonlinear image registration or brain tissue segmentation. This is made possible by our

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landmark-based diagnosis framework, which includes a two-layer shape constraint regression forest model to enhance the detection of AD with better efficiency and accuracy. In [9], various machine learning algorithms were employed for AD diagnosis. The results showed that Extreme Gradient Boosting, RF, and SVM achieved an accuracy of 89.63%. The classification algorithms such as K-Nearest Neighbors (KNN), DT, RF, NN, and SVM were utilized in diagnosing AD [10]. Moreover, different techniques for feature selection and dimension reduction were employed to identify more relevant features for decision-making, which in turn reduced the training time. The logistic regression and KNN classifier achieved an accuracy of nearly 90%.

The study utilized various classification algorithms to enhance the diagnostic process and improve the accuracy of detecting Alzheimer's Disease. Four specific algorithms were employed: XGBoost, AdaBoost, GradientBoost, and voting (ensemble) algorithms. Firstly, XGBoost, which stands for "Extreme Gradient Boosting," was one of the algorithms used in the study. It utilizes a gradient boosting framework to optimize the performance of classification models. Secondly, AdaBoost, or "Adaptive Boosting," was also utilized in the study. This algorithm focuses on iteratively boosting weak classifiers by assigning higher weights to misclassified instances, thus improving their predictive power. Next, the study employed the GradientBoost algorithm. This algorithm, similar to XGBoost, utilizes gradient boosting techniques to iteratively build strong classifiers by minimizing a loss function. Lastly, the study utilized voting (ensemble) algorithms. These algorithms combine the predictions made by multiple classifiers and make decisions based on majority voting. This ensemble approach helps enhance the overall performance and accuracy of the classification process. To evaluate the effectiveness of these algorithms, several metrics were employed. The obtained results showed that XGBoost achieved a classification accuracy of 85%, AdaBoost achieved a classification accuracy of 75%, GradientBoost achieved a classification accuracy of 85%, and the voting algorithm achieved a classification accuracy of 80%. These accuracy percentages represent the proportion of correctly classified instances in the dataset. Overall, the study demonstrated the efficacy of utilizing these classification algorithms, particularly XGBoost, GradientBoost, and voting algorithms, in enhancing the diagnostic process and improving the accuracy of detecting Alzheimer's Disease.

II. MATERIAL AND METHODS

A. Dataset description

This dataset is obtained from the machine learning repository from [11]. The dataset section of this academic paper focuses on the DARWIN dataset, which comprises handwriting information collected through the acquisition protocol. This protocol was purposely developed to detect Alzheimer's disease (AD) at an early stage and covers a total of 25 handwriting tasks. The dataset includes information on 174 participants, with 89 diagnosed with AD and 85 being healthy individuals [12-13].

B. METHODS

In this study, boosting classifier models, namely XGBoost, GradientBoost, and AdaBoost, were trained and tested using the features derived from the handwriting data specifically designed for Alzheimer's disease. These classifiers were then incorporated into an ensemble model voting classifier. The output of the voting classifier provided classifications for both healthy individuals and those diagnosed with Alzheimer's Disease. The division of the dataset into training, validation, and test sets is crucial for creating classification models that are reliable and can be applied to new data. This division helps to optimize the model's performance and evaluate its capabilities on unseen data. The division of the dataset is performed using the 5-Fold Cross Validation (FCV) method. AD classification flowchart is given in Fig. 1. One popular way to implement boosting is through a voting classifier pool. In this approach, multiple weak classifiers are trained independently, and their collective predictions are used to make the final classification decision. Each weak classifier assigns a weight to its prediction, and the final prediction is determined based on the cumulative weights of all the weak classifiers. XGBoost is a boosting algorithm that uses gradient boosting techniques to create an ensemble of DT. It focuses on optimizing the loss function during the training process and handles missing values and regularization to prevent overfitting. XGBoost has gained popularity due to its ability to handle large datasets efficiently and deliver accurate and robust predictions [14]. GradientBoost, as the name suggests, utilizes gradient descent optimization to iteratively build an ensemble of weak models. It aims to minimize a differentiable loss function, such as the mean squared error or log loss. By training subsequent models on the negative gradients of the loss function with respect to the previous model's predictions, GradientBoost can improve upon its predecessors and enhance the overall classification performance. Overall, Gradient Boosting is an iterative procedure that enhances the efficiency of the ensemble by acquiring knowledge from the errors committed by the prior models [15]. AdaBoost is another boosting algorithm that assigns weights to training instances based on their classification errors. It trains subsequent weak models on the instances that were misclassified by the previous models, paying more attention to these difficult instances. The final categorization is established through the weighted votes of the less effective classifiers, with additional emphasis placed on the more precise models [16]. The use of a voting classifier pool with XGBoost, GradientBoost, and AdaBoost enhances the ensemble model's predictive power by combining the strengths of each individual classifier. It allows for more robust and accurate classification, making it a valuable tool in various academic and practical domains. The acquisition protocol for the online handwriting data collection aimed at diagnosing AD was implemented. The dataset was meticulously gathered using a comprehensive protocol, consisting of 25 distinct tasks specifically chosen to elicit various aspects of the disease. Every data point in the dataset represents the actions performed during the completion of a task and is defined by a series of 18 features. These features encompass both those previously utilized by other researchers to study Parkinson's disease, as well as novel ones derived from neuroscience studies. The classification results are evaluated using various performance

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metrics, including accuracy, precision, recall, f-measures, and the Matthews Correlation Coefficient (MCC).

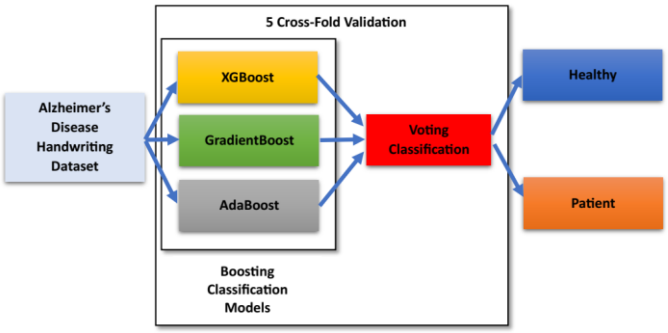


Fig. 1. Alzheimer's Disease classification flowchart

III. THE SIMULATION RESULTS AND DISCUSSION

In the study at hand, machine learning algorithms such as XGBoost, GradientBoost, AdaBoost, and voting were employed to determine the presence of Alzheimer's Disease. To evaluate the performance of these algorithms, a commonly used performance evaluation metric called the confusion matrix was utilized. Each algorithm was assessed individually, and the resulting confusion matrices were presented in the form of Fig. s. Fig. 2 displayed the confusion matrix for XGBoost, illustrating the distribution of predictions made by this algorithm. It showcased the number of correctly classified instances as well as the number of instances misclassified for the detection of Alzheimer's Disease. Similarly, Fig. 3 exhibited the confusion matrix for GradientBoost algorithm. This matrix provided a visual representation of the algorithm's performance. Lastly, Fig. 4 presented the confusion matrix for AdaBoost algorithm.

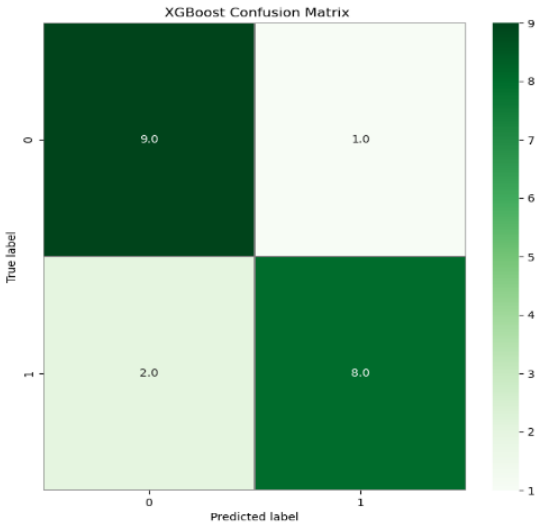


Fig. 2. XGBoost Confusion Matrix

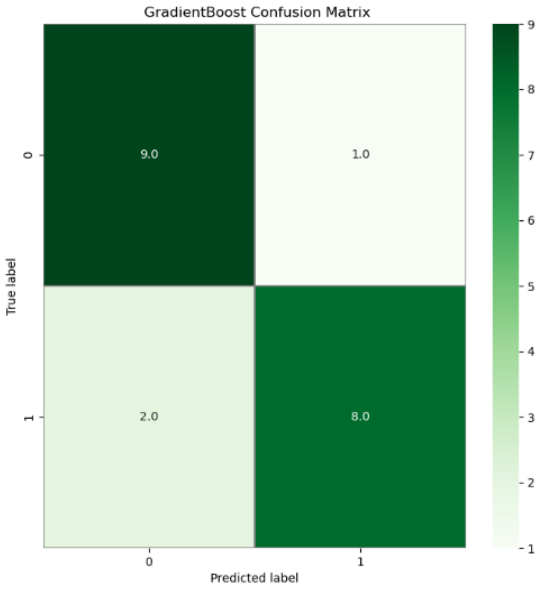


Fig. 3. GradientBoost Confusion Matrix

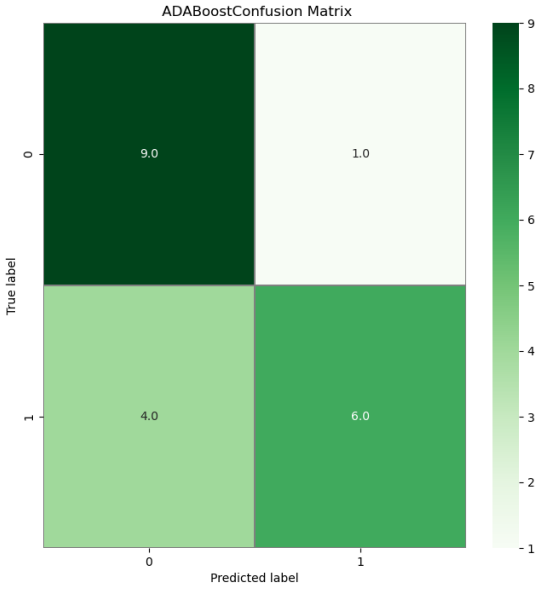


Fig. 4. AdaBoost Confusion Matrix

The efficacy of a classification model lies in its ability to predict the correct labels for new and unseen data points accurately, known as classification performance. To evaluate this performance, various performance metrics and equations can be utilized. These metrics and equations offer a comprehensive assessment of the model's classification performance, enabling researchers and practitioners to assess its accuracy, precision, recall, f-measure, and MCC, and make informed decisions regarding its effectiveness in predicting the

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correct labels for new and unseen data points. The classification performance has been assessed using the following equations:

$$\text{Classification Accuracy (\%)} = \frac{\mathbf{TP+TN}}{\mathbf{TP+FP+TN+FN}} \times \mathbf{100} \quad (1)$$

$$\text{Precision} = \frac{\mathbf{TP}}{\mathbf{TP+FP}} \quad (2)$$

$$\text{Recall} = \frac{\mathbf{TP}}{\mathbf{TP+FN}} \quad (3)$$

$$\text{f-measure} = \frac{\mathbf{2\times precision\times recall}}{\mathbf{precision+recall}} \quad (4)$$

$$\text{Matthews Correlation Coefficient (MCC)} = \frac{(\mathbf{TP\times TN})-(\mathbf{FP\times FN})}{\sqrt{(\boldsymbol{TP+FP})\times(\mathbf{TP+FN})\times(\mathbf{TN+FP})\times(\mathbf{TN+FN})}} \quad (5)$$

TABLE 1: Performance results for the XGBoost algorithm with 5-FCV

Classification Accuracy (%)	85.00
Precision	85.35
Recall	85.00
f-measure	84.96
MCC	70.35

TABLE 2: Performance results for the ADABoost algorithm with 5-FCV

Classification Accuracy (%)	75.00
Precision	77.40
Recall	75.00
f-measure	74.42
MCC	52.41

TABLE 3: Performance results for the GradientBoost algorithm with 5-FCV

Classification Accuracy (%)	85.00
Precision	85.35
Recall	85.00
f-measure	84.96
MCC	70.35

TABLE 4: Performance results for the voting algorithm with 5-FCV

Classification Accuracy (%)	80.00
Precision	81.25
Recall	80.00
f-measure	79.80
MCC	61.24

In this study, four different algorithms were evaluated for their performance using 5-Fold Cross Validation. Specifically, Table 1 displays the performance results for the XGBoost algorithm, while Table 2 presents the performance results for the ADABoost algorithm. Moving on to Table 3, the performance results for the GradientBoost algorithm can be found, and lastly, Table 4 showcases the performance results for the voting (ensemble) approach. Among these classifiers, both XGBoost and GradientBoost demonstrated exceptional performance, achieving an impressive accuracy rate of 85%. These results imply that both algorithms are highly effective in accurately predicting the target variable based on the given dataset.

IV. CONCLUSIONS

In this study, the classification performance of four different algorithms – XGBoost, GradientBoost, AdaBoost, and voting classification algorithms – was evaluated using a 5-fold cross-validation (5-FCV) approach. Each algorithm was tested on a dataset acquired from the machine learning repository of the University of California. The results of the study showed that XGBoost achieved a classification accuracy of 85% using the 5-FCV approach. This means that the algorithm correctly classified 85% of the instances in the dataset. On the other hand, ADABoost achieved a classification accuracy of 75%, indicating that it correctly classified 75% of the instances. GradientBoost achieved a classification accuracy of 85%, the same as XGBoost. This suggests that both algorithms performed equally well in correctly classifying instances. Lastly, the voting classification algorithms achieved a classification accuracy of 80% using the 5-FCV approach. This means that the ensemble of voting classification algorithms correctly classified 80% of the instances in the dataset. These results indicate that XGBoost and GradientBoost performed better than ADABoost and the voting classification algorithms in terms of classification accuracy. However, it is important to note that classification accuracy alone may not provide a complete understanding of the algorithm's performance, and other performance metrics such as precision, recall, f-measure, and MCC should also be considered when evaluating the performance of classification algorithms.

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